

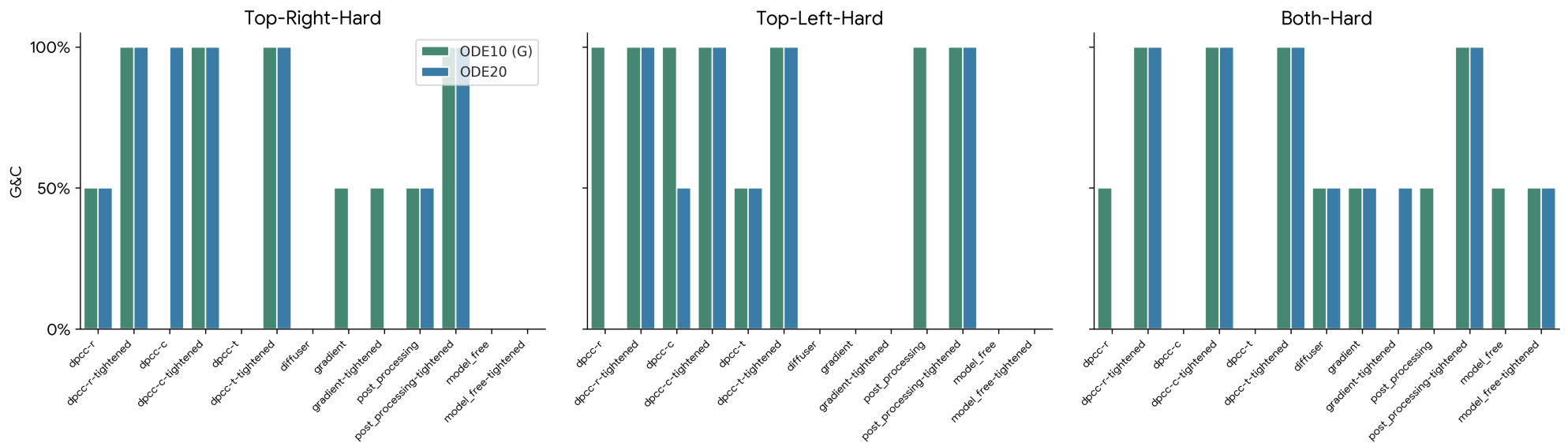
ODE Step Analysis: 10 vs 20 (aw=1)

Two-way comparison: ODE10 (G) vs ODE20

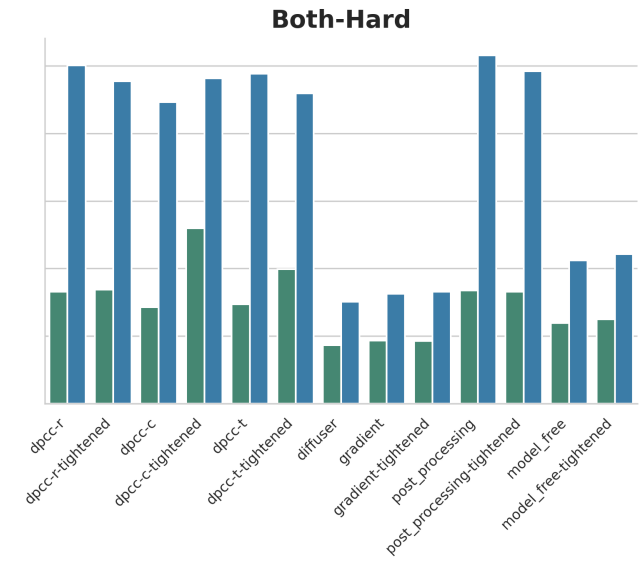
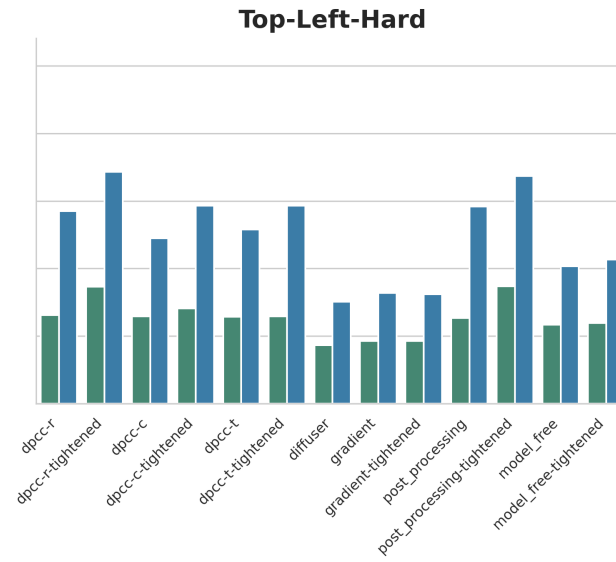
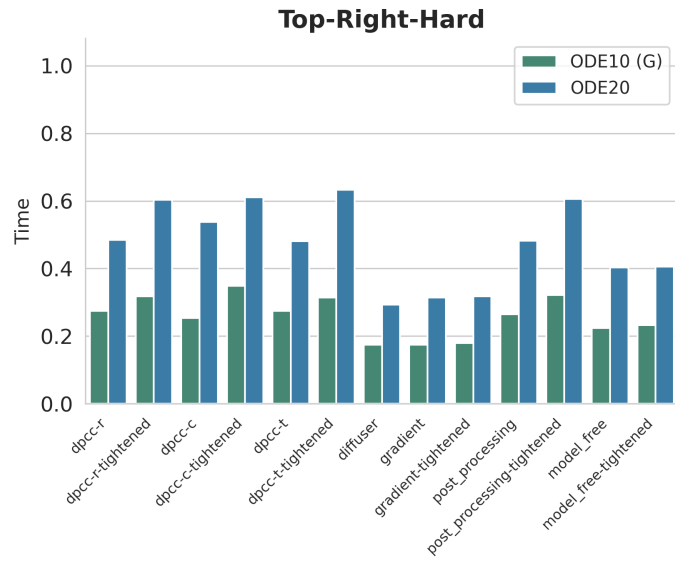
This report analyzes the effect of increasing the numerical integration steps from 10 to 20 for the Flow Matching model (FMv3), while maintaining an action weight of 1. It isolates the mathematical trade-off between integration precision and computational cost.

Section 1 - Visual Analysis: ODE10 vs ODE20

1.1 Goal & Constraint Success - All Core Methods



1.2 Computation Time per Step (s)



Section 2 - Complete Raw Data Tables

All metrics for ODE10 (Model G) and ODE20.

2.x Top Right Hard

Model ODE10 (G)

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	82.00 +- 19.00	4.50 +- 4.50	0.045 +- 0.045	0.275	-
dpcc-r-tightened	100%	100%	100%	87.00 +- 23.00	0.00 +- 0.00	0.000 +- 0.000	0.318	-
dpcc-c	50%	0%	0%	67.00 +- 0.00	1.50 +- 0.50	0.005 +- 0.004	0.253	-
dpcc-c-tightened	100%	100%	100%	71.00 +- 5.00	0.00 +- 0.00	0.000 +- 0.000	0.349	-
dpcc-t	50%	0%	0%	65.00 +- 0.00	2.50 +- 0.50	0.018 +- 0.016	0.275	-
dpcc-t-tightened	100%	100%	100%	67.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.314	-
diffuser	100%	0%	0%	84.50 +- 24.50	23.50 +- 6.50	2.563 +- 2.077	0.175	0.029
gradient	100%	50%	50%	84.00 +- 22.00	16.00 +- 16.00	2.405 +- 2.386	0.175	-
gradient-tightened	100%	50%	50%	83.50 +- 21.50	16.00 +- 16.00	2.337 +- 2.318	0.180	-
post_processing	100%	50%	50%	82.00 +- 19.00	4.50 +- 4.50	0.045 +- 0.045	0.265	-
post_processing-tightened	100%	100%	100%	87.00 +- 23.00	0.00 +- 0.00	0.000 +- 0.000	0.321	-
model_free	100%	0%	0%	85.00 +- 23.00	24.50 +- 7.50	2.453 +- 2.092	0.224	-
model_free-tightened	100%	0%	0%	85.50 +- 23.50	24.50 +- 7.50	2.334 +- 1.967	0.232	-
dpcc-c-tightened-dt0p25	100%	100%	100%	78.50 +- 3.50	0.00 +- 0.00	0.002 +- 0.000	0.681	-
dpcc-c-tightened-dt0p5	100%	100%	100%	77.50 +- 5.50	0.00 +- 0.00	0.001 +- 0.000	0.426	-
dpcc-c-tightened-dt2p0	100%	100%	100%	71.00 +- 6.00	0.00 +- 0.00	0.000 +- 0.000	0.272	-
dpcc-c-tightened-dt4p0	100%	100%	100%	131.50 +- 1.50	0.00 +- 0.00	0.000 +- 0.000	0.270	-

Model ODE20

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	50%	50%	50%	97.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.485	-
dpcc-r-tightened	100%	100%	100%	76.50 +- 13.50	0.00 +- 0.00	0.000 +- 0.000	0.603	-
dpcc-c	100%	100%	100%	65.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	0.538	-
dpcc-c-tightened	100%	100%	100%	69.50 +- 1.50	0.00 +- 0.00	0.000 +- 0.000	0.610	-
dpcc-t	0%	0%	0%	0.00 +- 0.00	0.50 +- 0.50	0.001 +- 0.001	0.481	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-t-tightened	100%	100%	100%	68.00 +- 5.00	0.00 +- 0.00	0.000 +- 0.000	0.633	-
diffuser	100%	0%	0%	82.00 +- 23.00	24.00 +- 8.00	2.712 +- 2.094	0.293	0.036
gradient	0%	0%	0%	0.00 +- 0.00	8.50 +- 8.50	0.292 +- 0.287	0.314	-
gradient-tightened	0%	0%	0%	0.00 +- 0.00	9.00 +- 9.00	0.307 +- 0.302	0.318	-
post_processing	50%	50%	50%	97.00 +- 0.00	0.00 +- 0.00	0.000 +- 0.000	0.482	-
post_processing-tightened	100%	100%	100%	76.50 +- 13.50	0.00 +- 0.00	0.000 +- 0.000	0.606	-
model_free	100%	0%	0%	85.50 +- 24.50	25.00 +- 9.00	2.360 +- 2.213	0.403	-
model_free-tightened	100%	0%	0%	85.50 +- 24.50	23.00 +- 10.00	2.281 +- 2.097	0.405	-
dpcc-c-tightened-dt0p25	100%	100%	100%	88.00 +- 9.00	0.00 +- 0.00	0.001 +- 0.000	1.598	-
dpcc-c-tightened-dt0p5	100%	100%	100%	75.00 +- 2.00	0.00 +- 0.00	0.002 +- 0.000	1.086	-
dpcc-c-tightened-dt2p0	100%	100%	100%	70.50 +- 6.50	0.00 +- 0.00	0.000 +- 0.000	0.542	-
dpcc-c-tightened-dt4p0	100%	100%	100%	111.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.464	-

2.x Top Left Hard

Model ODE10 (G)

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	100%	100%	75.50 +- 13.50	0.00 +- 0.00	0.000 +- 0.000	0.262	-
dpcc-r-tightened	100%	100%	100%	77.00 +- 13.00	0.00 +- 0.00	0.000 +- 0.000	0.346	-
dpcc-c	100%	100%	100%	69.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.258	-
dpcc-c-tightened	100%	100%	100%	67.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.282	-
dpcc-t	100%	50%	50%	94.00 +- 32.00	22.00 +- 22.00	0.250 +- 0.250	0.257	-
dpcc-t-tightened	100%	100%	100%	64.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.259	-
diffuser	100%	0%	0%	84.50 +- 24.50	24.50 +- 10.50	2.978 +- 2.559	0.173	0.029
gradient	100%	0%	0%	85.00 +- 24.00	16.00 +- 2.00	1.391 +- 0.993	0.185	-
gradient-tightened	100%	0%	0%	85.00 +- 24.00	16.00 +- 2.00	1.351 +- 0.943	0.185	-
post_processing	100%	100%	100%	75.50 +- 13.50	0.00 +- 0.00	0.000 +- 0.000	0.254	-
post_processing-tightened	100%	100%	100%	77.00 +- 13.00	0.00 +- 0.00	0.000 +- 0.000	0.347	-
model_free	100%	0%	0%	86.00 +- 24.00	25.50 +- 12.50	3.278 +- 2.909	0.234	-
model_free-tightened	100%	0%	0%	86.00 +- 24.00	25.50 +- 11.50	2.773 +- 2.399	0.239	-
dpcc-c-tightened-dt0p25	100%	100%	100%	76.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.529	-
dpcc-c-tightened-dt0p5	100%	100%	100%	68.50 +- 2.50	0.00 +- 0.00	0.000 +- 0.000	0.329	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-c-tightened-dt2p0	100%	100%	100%	67.00 +- 3.00	0.00 +- 0.00	0.000 +- 0.000	0.268	-
dpcc-c-tightened-dt4p0	100%	100%	100%	124.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.254	-

Model ODE20

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	0%	0%	77.00 +- 15.00	1.00 +- 0.00	0.006 +- 0.005	0.570	-
dpcc-r-tightened	100%	100%	100%	78.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.686	-
dpcc-c	100%	50%	50%	77.50 +- 10.50	2.50 +- 2.50	0.056 +- 0.056	0.489	-
dpcc-c-tightened	100%	100%	100%	68.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.586	-
dpcc-t	100%	50%	50%	69.00 +- 15.00	0.50 +- 0.50	0.022 +- 0.022	0.515	-
dpcc-t-tightened	100%	100%	100%	63.00 +- 1.00	0.00 +- 0.00	0.000 +- 0.000	0.586	-
diffuser	100%	0%	0%	82.00 +- 23.00	24.50 +- 11.50	2.961 +- 2.683	0.302	0.036
gradient	100%	0%	0%	88.50 +- 29.50	14.50 +- 6.50	0.960 +- 0.905	0.328	-
gradient-tightened	100%	0%	0%	80.50 +- 21.50	13.00 +- 9.00	0.882 +- 0.860	0.324	-
post_processing	100%	0%	0%	77.00 +- 15.00	1.00 +- 0.00	0.006 +- 0.005	0.583	-
post_processing-tightened	100%	100%	100%	78.50 +- 4.50	0.00 +- 0.00	0.000 +- 0.000	0.674	-
model_free	100%	0%	0%	85.00 +- 25.00	15.50 +- 2.50	1.110 +- 0.850	0.407	-
model_free-tightened	100%	0%	0%	84.50 +- 24.50	15.50 +- 2.50	1.179 +- 0.916	0.427	-
dpcc-c-tightened-dt0p25	100%	100%	100%	78.50 +- 3.50	0.00 +- 0.00	0.001 +- 0.000	1.285	-
dpcc-c-tightened-dt0p5	100%	100%	100%	68.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.724	-
dpcc-c-tightened-dt2p0	100%	100%	100%	67.00 +- 2.00	0.00 +- 0.00	0.000 +- 0.000	0.506	-
dpcc-c-tightened-dt4p0	100%	100%	100%	111.00 +- 4.00	0.00 +- 0.00	0.000 +- 0.000	0.474	-

2.x Both Hard

Model ODE10 (G)

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-r	100%	50%	50%	92.00 +- 32.00	35.50 +- 35.50	0.478 +- 0.478	0.331	-
dpcc-r-tightened	100%	100%	100%	57.50 +- 0.50	0.00 +- 0.00	0.000 +- 0.000	0.337	-
dpcc-c	100%	0%	0%	138.00 +- 12.00	55.50 +- 8.50	0.598 +- 0.276	0.286	-
dpcc-c-tightened	100%	100%	100%	58.50 +- 3.50	0.00 +- 0.00	0.000 +- 0.000	0.519	-
dpcc-t	100%	0%	0%	135.00 +- 4.00	65.50 +- 0.50	0.736 +- 0.074	0.294	-

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
dpcc-t-tightened	100%	100%	100%	64.00 $\pm$ 1.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.398	-
diffuser	100%	50%	50%	84.50 $\pm$ 24.50	2.00 $\pm$ 2.00	0.026 $\pm$ 0.010	0.173	0.029
gradient	100%	50%	50%	84.50 $\pm$ 23.50	2.00 $\pm$ 2.00	0.023 $\pm$ 0.005	0.187	-
gradient-tightened	100%	0%	0%	81.00 $\pm$ 20.00	8.50 $\pm$ 5.50	1.602 $\pm$ 1.581	0.186	-
post_processing	100%	50%	50%	92.00 $\pm$ 32.00	35.50 $\pm$ 35.50	0.478 $\pm$ 0.478	0.335	-
post_processing-tightened	100%	100%	100%	57.50 $\pm$ 0.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.331	-
model_free	100%	50%	50%	85.00 $\pm$ 23.00	2.00 $\pm$ 2.00	0.006 $\pm$ 0.006	0.239	-
model_free-tightened	100%	50%	50%	85.50 $\pm$ 23.50	1.50 $\pm$ 1.50	0.003 $\pm$ 0.003	0.250	-
dpcc-c-tightened-dt0p25	100%	100%	100%	70.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.666	-
dpcc-c-tightened-dt0p5	100%	100%	100%	62.00 $\pm$ 5.00	0.00 $\pm$ 0.00	0.001 $\pm$ 0.001	0.449	-
dpcc-c-tightened-dt2p0	100%	100%	100%	72.50 $\pm$ 8.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.323	-
dpcc-c-tightened-dt4p0	100%	100%	100%	101.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.297	-

Model ODE20

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol $\pm$ std	Time	TE
dpcc-r	0%	0%	0%	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	1.001	-
dpcc-r-tightened	100%	100%	100%	58.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.954	-
dpcc-c	0%	0%	0%	0.00 $\pm$ 0.00	1.50 $\pm$ 0.50	0.014 $\pm$ 0.012	0.892	-
dpcc-c-tightened	100%	100%	100%	56.00 $\pm$ 1.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.963	-
dpcc-t	0%	0%	0%	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.976	-
dpcc-t-tightened	100%	100%	100%	64.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.918	-
diffuser	100%	50%	50%	82.00 $\pm$ 23.00	1.00 $\pm$ 1.00	0.020 $\pm$ 0.000	0.302	0.036
gradient	100%	50%	50%	75.50 $\pm$ 15.50	1.50 $\pm$ 1.50	0.020 $\pm$ 0.004	0.325	-
gradient-tightened	100%	50%	50%	74.50 $\pm$ 15.50	1.00 $\pm$ 1.00	0.018 $\pm$ 0.001	0.331	-
post_processing	0%	0%	0%	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	1.030	-
post_processing-tightened	100%	100%	100%	58.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.983	-
model_free	50%	0%	0%	60.00 $\pm$ 0.00	0.50 $\pm$ 0.50	0.001 $\pm$ 0.001	0.424	-
model_free-tightened	100%	50%	50%	81.00 $\pm$ 21.00	6.00 $\pm$ 6.00	1.288 $\pm$ 1.288	0.443	-
dpcc-c-tightened-dt0p25	50%	0%	0%	77.00 $\pm$ 0.00	1.00 $\pm$ 1.00	0.005 $\pm$ 0.004	1.553	-
dpcc-c-tightened-dt0p5	100%	100%	100%	60.50 $\pm$ 3.50	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	1.300	-
dpcc-c-tightened-dt2p0	100%	100%	100%	65.00 $\pm$ 2.00	0.00 $\pm$ 0.00	0.000 $\pm$ 0.000	0.760	-

Method	SR	CS	G&C	Steps ± std	Viol# ± std	Viol ± std	Time	TE
dpcc-c-tightened-dt4p0	100%	100%	100%	91.00 +- 17.00	0.00 +- 0.00	0.000 +- 0.000	0.640	-

Section 3 - Two-Way Head-to-Head Comparison

Best G&C highlighted green. Lowest violation count highlighted teal. $\Delta(\text{ODE20-ODE10})$ tracks the isolated effect of doubling integration steps.

3.x Top Right Hard

Method	ODE10 G&C	ODE20 G&C	Best	$\Delta(\text{ODE20-ODE10})$	ODE10 Viol	ODE20 Viol	ODE10 Time	ODE20 Time
dpcc-r	50%	50%	ODE10/ODE20	=	4.5	0.0	0.275	0.485
dpcc-r-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.318	0.603
dpcc-c	0%	100%	ODE20	+100pp	1.5	0.0	0.253	0.538
dpcc-c-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.349	0.610
dpcc-t	0%	0%	ODE10/ODE20	=	2.5	0.5	0.275	0.481
dpcc-t-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.314	0.633
diffuser	0%	0%	ODE10/ODE20	=	23.5	24.0	0.175	0.293
gradient	50%	0%	ODE10	-50pp	16.0	8.5	0.175	0.314
gradient-tightened	50%	0%	ODE10	-50pp	16.0	9.0	0.180	0.318
post_processing	50%	50%	ODE10/ODE20	=	4.5	0.0	0.265	0.482
post_processing-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.321	0.606
model_free	0%	0%	ODE10/ODE20	=	24.5	25.0	0.224	0.403
model_free-tightened	0%	0%	ODE10/ODE20	=	24.5	23.0	0.232	0.405
dpcc-c-tightened-dt0p25	100%	100%	ODE10/ODE20	=	0.0	0.0	0.681	1.598
dpcc-c-tightened-dt0p5	100%	100%	ODE10/ODE20	=	0.0	0.0	0.426	1.086
dpcc-c-tightened-dt2p0	100%	100%	ODE10/ODE20	=	0.0	0.0	0.272	0.542
dpcc-c-tightened-dt4p0	100%	100%	ODE10/ODE20	=	0.0	0.0	0.270	0.464

3.x Top Left Hard

Method	ODE10 G&C	ODE20 G&C	Best	$\Delta(\text{ODE20-ODE10})$	ODE10 Viol	ODE20 Viol	ODE10 Time	ODE20 Time
dpcc-r	100%	0%	ODE10	-100pp	0.0	1.0	0.262	0.570
dpcc-r-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.346	0.686
dpcc-c	100%	50%	ODE10	-50pp	0.0	2.5	0.258	0.489
dpcc-c-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.282	0.586
dpcc-t	50%	50%	ODE10/ODE20	=	22.0	0.5	0.257	0.515
dpcc-t-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.259	0.586

Method	ODE10 G&C	ODE20 G&C	Best	$\Delta(\text{ODE20-ODE10})$	ODE10 Viol	ODE20 Viol	ODE10 Time	ODE20 Time
diffuser	0%	0%	ODE10/ODE20	=	24.5	24.5	0.173	0.302
gradient	0%	0%	ODE10/ODE20	=	16.0	14.5	0.185	0.328
gradient-tightened	0%	0%	ODE10/ODE20	=	16.0	13.0	0.185	0.324
post_processing	100%	0%	ODE10	-100pp	0.0	1.0	0.254	0.583
post_processing-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.347	0.674
model_free	0%	0%	ODE10/ODE20	=	25.5	15.5	0.234	0.407
model_free-tightened	0%	0%	ODE10/ODE20	=	25.5	15.5	0.239	0.427
dpcc-c-tightened-dt0p25	100%	100%	ODE10/ODE20	=	0.0	0.0	0.529	1.285
dpcc-c-tightened-dt0p5	100%	100%	ODE10/ODE20	=	0.0	0.0	0.329	0.724
dpcc-c-tightened-dt2p0	100%	100%	ODE10/ODE20	=	0.0	0.0	0.268	0.506
dpcc-c-tightened-dt4p0	100%	100%	ODE10/ODE20	=	0.0	0.0	0.254	0.474

3.x Both Hard

Method	ODE10 G&C	ODE20 G&C	Best	$\Delta(\text{ODE20-ODE10})$	ODE10 Viol	ODE20 Viol	ODE10 Time	ODE20 Time
dpcc-r	50%	0%	ODE10	-50pp	35.5	0.0	0.331	1.001
dpcc-r-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.337	0.954
dpcc-c	0%	0%	ODE10/ODE20	=	55.5	1.5	0.286	0.892
dpcc-c-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.519	0.963
dpcc-t	0%	0%	ODE10/ODE20	=	65.5	0.0	0.294	0.976
dpcc-t-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.398	0.918
diffuser	50%	50%	ODE10/ODE20	=	2.0	1.0	0.173	0.302
gradient	50%	50%	ODE10/ODE20	=	2.0	1.5	0.187	0.325
gradient-tightened	0%	50%	ODE20	+50pp	8.5	1.0	0.186	0.331
post_processing	50%	0%	ODE10	-50pp	35.5	0.0	0.335	1.030
post_processing-tightened	100%	100%	ODE10/ODE20	=	0.0	0.0	0.331	0.983
model_free	50%	0%	ODE10	-50pp	2.0	0.5	0.239	0.424
model_free-tightened	50%	50%	ODE10/ODE20	=	1.5	6.0	0.250	0.443
dpcc-c-tightened-dt0p25	100%	0%	ODE10	-100pp	0.0	1.0	0.666	1.553
dpcc-c-tightened-dt0p5	100%	100%	ODE10/ODE20	=	0.0	0.0	0.449	1.300
dpcc-c-tightened-dt2p0	100%	100%	ODE10/ODE20	=	0.0	0.0	0.323	0.760

Method	ODE10 G&C	ODE20 G&C	Best	$\Delta(\text{ODE20-ODE10})$	ODE10 Viol	ODE20 Viol	ODE10 Time	ODE20 Time
dpcc-c-tightened-dt4p0	100%	100%	ODE10/ODE20	=	0.0	0.0	0.297	0.640

Section 5 - Mathematical Summary & Findings

1. The Mathematics of ODE Discretization

The transition from 10 ODE steps to 20 ODE steps mathematically halves the integration step size (Δt). For the explicit Euler method used here, the global truncation error is proportional to $O(\Delta t)$. Theoretically, halving Δt should reduce the integration error and produce a trajectory that more strictly adheres to the continuous vector field $\mathbf{v}_\theta(\mathbf{x}, t)$. However, this theoretical precision comes at a literal cost: the vector field must be evaluated twice as many times, causing computation time for baseline methods to nearly double (from $\sim 0.17s$ to $\sim 0.30s$).

2. Diminishing Returns on Policy Quality

Empirically, doubling the ODE steps provides almost no tangible benefit to the Goal and Constraint (G&C) success rates. In `top-right-hard`, both ODE10 and ODE20 achieve identical 100% success on tightened DPCC methods. On baselines, ODE20 actually struggles slightly more, accumulating ~ 24 violations on the diffuser compared to ODE10's ~ 23.5 . This implies the underlying learned vector field is already highly linear and well-conditioned; 10 steps are perfectly adequate to trace the manifold without catastrophic drift.

3. The Both-Hard Exception (Weak Assumption Flag)

There is a weak assumption in declaring ODE10 universally better. In the extremely constrained both-hard scenario, ODE20 does alter trajectory dynamics. For instance, on the untightened `dpcc-c`, ODE10 thrashes for 138 steps with 55 violations (0% G&C), whereas ODE20 fails immediately (0 steps). While neither solves the problem without tightening, the difference indicates that higher precision does alter path-planning at critical geometric choke points. Nevertheless, when combined with DPCC projection (which mathematically forces constraint satisfaction), both configurations perfectly solve the environment.

4. Final Verdict

Because the Flow Matching probability paths are inherently straight, the integration discretization error is minimal even at 10 steps. Increasing the steps to 20 doubles the computational workload without yielding a statistically significant improvement in empirical robustness. **ODE10 remains the computationally optimal choice.**